

PRE MANUFACTURE UNIT TEST CHECKLIST

This unit test checklist will serve as a valuable tool when commissioning your PCB.

Fill this in thoroughly. Please

It will help to understand the correct function of your PCB, as well as design errors will become clear.

PCB Design Intent	Completed:	YES	
	Reviewed:	YES	

Stage	Function	Description	Verification Method	Steps	Result1	Result2	Result3	Notes
Before Assembly	QUALITY CONTROL	PCB Quality Control	Inspection	Check the PCB is has been manufactured to our expected quality as well as all hardware	PASS	PASS	PASS	
Before Assembly	CORRECTLY ASSEMBLED	All Hardware Accountable	Inspection	Check the PCB and KICAD schematics to ensure all components are accounted for (before assembly due to A.C assembly)	PASS	PASS	PASS	
During Assembly - Unpowered	POWER DISTRIBUTION	Power and Ground are isolated	Test	Use a multimeter to test that the resistance between power and ground is an open loop	PASS	PASS	PASS	
During Assembly - Unpowered	Setup	Solder Header Pins	Test	Measure continuity of joints	PASS	PASS	PASS	
DISCHARGE								
During Assembly - Powered	LV Power	Connect 12V to LV side	Demonstration	DISCHARGE OKHS LED stays off	PASS	PASS	PASS	
During Assembly - Unpowered	Setup	Check relay polarity	Test	Cross check with past TSAL and use power supply to verify	PASS	PASS	PASS	
During Assembly - Unpowered	Setup	Solder DCDC & Relay	Inspection		PASS	PASS	PASS	
During Assembly - Unpowered	Setup	Solder PTC Thermistor & Dis Resistor	Inspection		PASS	PASS	PASS	
Post Assembly	LV Power	Connect 12V to LV side	Demonstration	DISCHARGE OKHS LED turns ON	PASS	PASS	PASS	
Post Assembly		Disconnect power and measure resistance between TS+ and TS-	Test	5k ohm measured	PASS	PASS	PASS	
Post Assembly	Shutdown power	Connect 12V to SHUTDOWN_IN, measure resistance between TS+ and TS-	Test	Open circuit measured	PASS	PASS	PASS	2M ohm?? -> same as old TSAL
Final Bench Test	LV Power	Connect LV to the LV side, Using heat gun, heat thermistor up.	Demonstration	DISCHARGE OKHS LED turns ON and then turns OFF when heat is applied		PASS	PASS	Board 1: Scrapped for Relay and iso 12V reg
Final Bench Test	HV Power	Connect TS+ and TS- to HV power supply. Turn power supply to 450V, and turn on for 3 seconds, using thermal camera measure temperature of resistor.	Demonstration	Resistor gets hot.		PASS	PASS	
Final Bench Test	HV Power	Connect 12V to SHUTDOWN_IN, and turn HV Supply on	Demonstration	Resistor doesn't get hot, and no current is drawn from power supply		PASS	PASS	
TSAL								
Final Bench Test	LV Power	(PRE DCDC solder) Connect 12V to the LV side and GND to TSAL_RED	Demonstration	Red TSAL starts flashing		PASS	PASS	
Final Bench Test	LV Power	Connect 12V to the LV side and GND to TSAL_GREEN	Demonstration	Green TSAL is on	PASS	PASS	PASS	
Final Bench Test	HV Power	Connect power supply to HV side, With the HV Power still turned off turn on the LV Power supply.	Demonstration	Green TSAL is on	PASS	PASS	PASS	
Final Bench Test	HV Power	Set HV power supply up to 50V and turn it on, Turn on LV power supply.	Demonstration	Green TSAL is on	PASS	PASS	PASS	
Final Bench Test	HV Power	With the power supplies turned on, slowly increase the voltage of the HV power supply.	Demonstration	Red TSAL starts flashing when the voltage is about 60V	FAIL	PASS	PASS	Board 1: RED 0603 LED broken
Final Bench Test	HV Power	Turn HV power supply up and down, ensuring LED changes from Green and Red when it should	Demonstration	Green and Red turn on when required	FAIL	PASS	PASS	Board 1: Scrapped for Relay and iso 12V reg
Integration Test	TSAL Car Test	Connect TSAL_GREEN_SWITCHED_GND to the TSAL. Turn on LV Power.	Demonstration	The green LED on the TSAL should turn on		PASS		
Integration Test	TSAL Car Test	Set HV voltage to 50V, Turn HV power supply ON and increase voltage.	Demonstration	The green LED turns off when the HV approaches 60V		PASS		
Integration Test	TSAL Car Test	Turn HV Power supply off. Connect TSAL_RED_SWITCHED_GND to the TSAL. Turn on LV Power.	Demonstration	Both LEDs on the TSAL should be OFF		PASS		
Integration Test	TSAL Car Test	Set HV voltage to 50V, Turn HV power supply ON and increase voltage.	Demonstration	The Red LED on the TSAL starts flashing when the HV approaches 60V		PASS		